
```

close all

load('materials/references_06.mat');
%Initialize values
m=0.5;
L=0.25;
k=3*10^-6;
b=10^-7;
g=9.81;
kd=0.25;
Ixx=5*10^-3;
Iyy=5*10^-3;
Izz=10^-2;
cm=10^4;

%Equilibrium
% Initialize variables
xdot_e = 0;
ydot_e = 0;
zdot_e = 0;
vxdot_e = 0;
vydot_e = 0;
vzdot_e = 0;
phidot_e = 0;
thetadot_e = 0;
psidot_e = 0;
omegaxdot_e = 0;
omegaydot_e = 0;
omegazdot_e = 0;

u_e = g*m/(4*k*cm);
u_e_sum = 4*u_e;

% Define A matrix
A = [...
    0 0 0 1 0 0 0 0 0 0 0 0 0;
    0 0 0 0 1 0 0 0 0 0 0 0 0;
    0 0 0 0 0 1 0 0 0 0 0 0 0;
    0 0 0 -kd/m 0 0 0 k*cm/m*u_e_sum 0 0 0 0 0;
    0 0 0 0 -kd/m 0 -k*cm/m*u_e_sum 0 0 0 0 0;
    0 0 0 0 0 -kd/m 0 0 0 0 0 0 0;
    0 0 0 0 0 0 0 0 0 0 1 0 0;
    0 0 0 0 0 0 0 0 0 0 0 1 0;
    0 0 0 0 0 0 0 0 0 0 0 0 1;
    0 0 0 0 0 0 0 0 0 0 0 0 0;
    0 0 0 0 0 0 0 0 0 0 0 0 0;
    0 0 0 0 0 0 0 0 0 0 0 0 0;
];

% Define B matrix
B = [

```

```

0 0 0 0; % 1
0 0 0 0; % 2
0 0 0 0; % 3
0 0 0 0; % 4
0 0 0 0; % 5
k*cm/m k*cm/m k*cm/m k*cm/m; % 6
0 0 0 0; % 7
0 0 0 0; % 8
0 0 0 0; % 9
L*k*cm/Ixx 0 -L*k*cm/Ixx 0; % 10
0 L*k*cm/Iyy 0 -L*k*cm/Iyy; % 11
b*cm/Izz -b*cm/Izz b*cm/Izz -b*cm/Izz; % 12
]

%Define C matrix
C = [...
1 0 0 0 0 0 0 0 0 0 0 0; % x
0 1 0 0 0 0 0 0 0 0 0 0; % y
0 0 1 0 0 0 0 0 0 0 0 0; % z
0 0 0 0 0 0 1 0 0 0 0 0; % roll (phi)
0 0 0 0 0 0 0 1 0 0 0 0; % pitch (theta)
0 0 0 0 0 0 0 0 1 0 0 0; % yaw (psi)
];

% Define the input
t = 0:0.01:5; % Time vector from 0 to 5 seconds with a step size
of 0.01 seconds
u = zeros(length(t), 4); % Initialize input matrix with the correct
dimensions

u(t >= 1 & t < 5, :) = 5; % Set input to [5 5 5 5] for 1s <= t < 5s
% Create the state-space model
sys = ss(A, B, C, []);

% Save system
save('lin_sys.mat', 'A', 'B', 'C', 'sys');

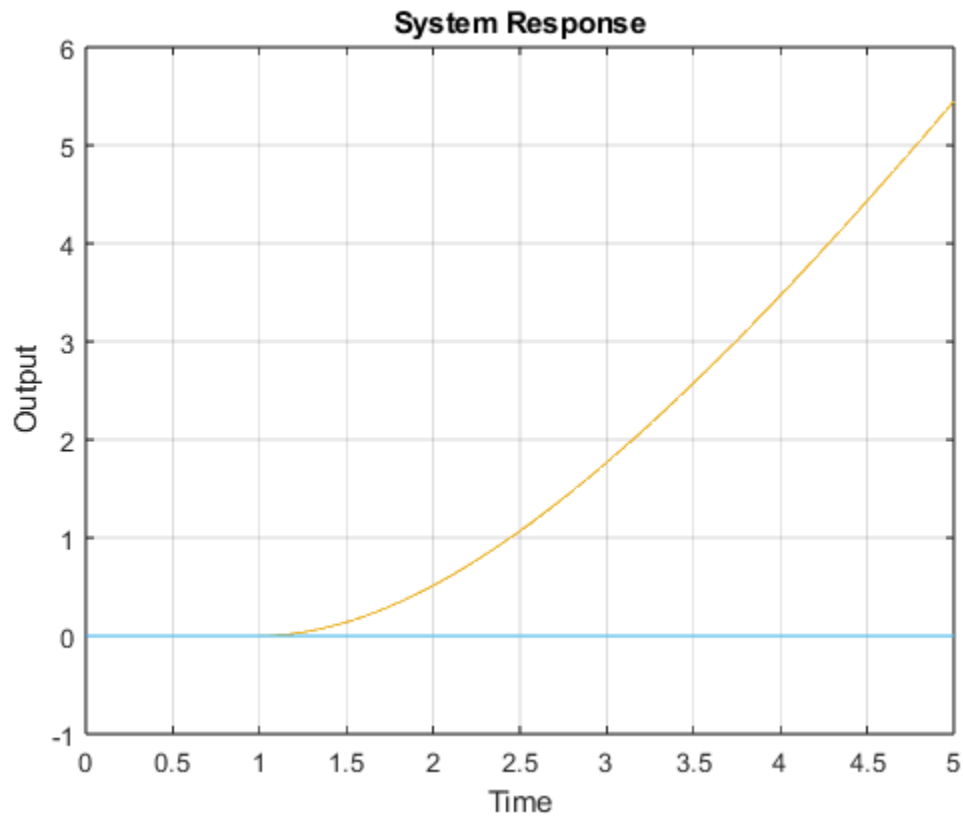
% Simulate the system
y = lsim(sys, u, t);
% Plot the output
plot(t, y);
grid on;
xlabel('Time');
ylabel('Output');
title('System Response');

B =

0 0 0 0
0 0 0 0
0 0 0 0
0 0 0 0
0 0 0 0

```

0.0600	0.0600	0.0600	0.0600
0	0	0	0
0	0	0	0
0	0	0	0
1.5000	0	-1.5000	0
0	1.5000	0	-1.5000
0.1000	-0.1000	0.1000	-0.1000



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